# ME 5309 – Mid-Semester Examination: ANSWER SHEET

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**Questions attempted:** Q1, Q2, Q4

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## Question 1

**(a)** (i) $f_x = f/p = 16/5 =$ **3.2 px**.

(ii) Width of sensor = 1280 × 5 = 6400. FOV = $N \cdot Z/f_x = 1280 \times 400 / 3.2 =$ **160000 mm**.

One pixel = FOV/N = 160000/1280 = **125 mm per pixel**. (looks large but the formula is right)

**(b)** LSB = 5/4096 = 1.2207 mV.

$\Delta F$ = LSB / S = 1.2207 mV / 10 mV/N = **0.0122 N**.

**(c)** $F = e^{AT}$ and $G = \int_0^T e^{A\tau} B \, d\tau$.

$F = e^{-0.2} =$ **0.8187**.

$G = \int_0^{0.1} e^{-2\tau} d\tau = 1 - e^{-0.2} =$ **0.1813**.

## Question 2

**(a)** Let $x_1 = \theta$ and $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau + b\, x_2 - mg\ell \sin x_1]$.

Equilibrum: $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Becuase $-1 \le \sin\theta \le 1$ we need $|\tau_0| \le mg\ell = 2.4525$ N·m.

**(b)** Jacobian at $\theta_e = 60°$: A = [0, 1; $-(mg\ell/J)\cos\theta_e$, $-b/J$].

$mg\ell/J = 2.4525/0.02 = 122.625$; $\cos(60) = -0.9524$, so $A_{21} = -122.625 \times (-0.9524) = +116.8$. $A_{22} = -b/J = -2.5$.

**A = [0, 1; 116.8, −2.5]**, **B = [0; 50]**.

**(c)** At rest $\dot{\theta} = \ddot{\theta} = 0$, so $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$, i.e. $50(90 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$.

Gravity torque is non-zero at 90°, so the left side can't be zero, hence $\theta_{ss} \neq \theta_d$.

Taking $\sin\theta_{ss} \approx 1$: $90 - \theta_{ss} = 2.4525/50 = 0.049$, so the stedy-state error ≈ **0.049°**.

**(d)** Error: $K_p(\theta_d - \theta_{ss}) = 0$ at rest, so $\theta_{ss} = \theta_d$; **error = 0**.

Pols: yes they change. The graviy term is removed from the dynamics so the effective stiffness changes from $K_p + mg\ell\cos\theta$ to $K_p$, which moves the poles (the system becomes faster).

## Question 4

**(a)** $\ddot{\theta} + a\dot{\theta} + c\theta = b(N_{ff}\, r - k_1\theta - k_2\dot{\theta}) \Rightarrow \ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\, r$.

With a = 2, b = 1, c = 6, $k_1 = k_2 = 2$: $s^2 + 4s + 8 = 0 \Rightarrow s = -2 \pm 2j$ (stable).

DC gain = $bN_{ff}/(c + bk_1) = N_{ff}/8 = 1 \Rightarrow$ **$N_{ff} = 8$**.

**(b)** $G_{DC} = bN_{ff}/(c + bk_1)$. Numerator: 0.75 × 8 = 6. Denominator: $c + k_1 = 8$ as before.

$\theta_{ss} = 6/8 =$ **0.75**, so error = **25%**.

**(c)** Char. poly: $s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 16s + 32$ (combining terms).

$2 + k_2 = 8 \Rightarrow$ **$k_2 = 6$**; $6 + k_1 = 16 \Rightarrow$ **$k_1 = 10$**; $K_I =$ **32**.

**(d)** In steady state $\dot{\xi} = 0$, so $r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$ for any b and c as long as the system is stabel.

Failure mode: integral action amplifies noise and makes the system oscillate. Remedy: put a low-pass filter on the measurement.